

Cadenza-Camerata Research Concert

Stoller Hall, 29th October 2026

Welcome to the Cadenza Camerata Research Concert

This pioneering event is a collaboration between the Cadenza Project and Manchester Camerata. Your answers will help us understand how to improve music for people with hearing loss and shape future research. Participating in the research activities is entirely voluntary, however.

You will hear the orchestra playing movements of Beethoven's 6th Symphony.

Schedule:

1) Complete the background questionnaire (pgs. 3-4) and have a look at the Research Activity instructions (pgs. 5-6).
2) Movement I – just sit back and enjoy.
3) Research Activity featuring 10 extracts from movements I and IV played in different ways. You will give feedback by the questionnaire contained within this booklet.
4) Movement V – sit back and enjoy or do the optional research task.
5) Q&A with members of the Cadenza Team on hearing loss and music.

We will make announcements and guide you through the research activities during the concert.

Many thanks for attending and please enjoy the event! The Cadenza Project team



(left to right: Jon Barker, Bruno Fazenda, Gerardo Doa Rabike, Michael Akeroyd, Jennifer Firth, Simone Graetzer (on the telly), Alinka Greasley, William Whitmer, Trevor Cox, Scott Bannister & Rebecca Vos)

1. Background Questionnaire

NOTE: By completing this research activity and handing in your booklet, you are giving your consent to these responses being used anonymously for purposes of research publication and for re-use by other researchers via data repositories. The study has been given ethical approval by the University of Leeds (Ref: AHC FREC 4477).

About You

Please complete the following information about you:

1. Seat row and number (we need to know where you are sitting when analysing your answers).

   Seat Row: _____ Seat Number: _____

2. Age (years): _____

3. Gender (please circle): Female Male Non-binary Prefer not to say

4. How familiar are you with Beethoven's 6th Symphony (Pastoral Symphony)? Please circle a number from 1-5.

not at all familiar	slightly familiar	somewhat familiar	very familiar	extremely familiar
1	**2**	**3**	**4**	**5**

5. If you are familiar at all with it, how much do you like Beethoven's 6th Symphony? Please circle a number from 0-10.

dislike extremely					neither like nor dislike					like extremely
0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

6. Are you a musician? Please circle which title best describes you:

Non-musician	**Music-loving non-musician**	**Amateur musician**	**Serious amateur musician**	**Semi-professional musician**	**Professional musician**

About your hearing

7. Without hearing assistance (e.g., hearing aids, implants), how much difficulty do you have in a conversation with one other person in a quiet place? Please circle which best describes your experience:

No difficulty	Only slight difficulty	Moderate difficulty	Great difficulty	Cannot manage at all

8. Please indicate (circle) whether you have the following hearing conditions:

Hearing loss (whether one or two ears): **Yes** **No** **Not sure**

Hyperacusis (heightened sensitivity to loud sounds): **Yes** **No** **Not sure**

Tinnitus (extended ringing in the ears when no external sound is present): **Yes** **No** **Not sure**

9. Do you wear hearing aid(s) or cochlear implant(s) devices? Please circle: **Yes** **No**

If yes, please circle what you are **currently wearing at this concert** in each ear:

Nothing: **Left Ear** **Right Ear**

Hearing aid: **Left Ear** **Right Ear**

Cochlear implant: **Left Ear** **Right Ear**

Bone-anchored aid: **Left Ear** **Right Ear**

10. If you are wearing hearing aid(s) or implant(s), do you currently have it set to a music program?

Please circle: **Yes** **No** **Don't know**

11. If you are wearing hearing aid(s) or implant(s), do you normally wear them to orchestral concerts?

Please circle: **Never** **Sometimes** **Always**

2. Definitions

Here are the attributes we will ask you to rate each of the extracts on. Please take a couple of minutes to familiarise yourself with these. A brief version of these definitions will be also provided next to the questions.

Clarity

Clarity refers to how well you can hear and distinguish between the different instruments and elements within the music. Unclear music may sound indistinct, mushy or muddy. Clear music may sound clean and distinct.

Distortion

Distortion can be caused by artefacts that should not be there or do not feel right, e.g., noise, hiss, pops or crackles. It can also be caused by pitches sounding wrong. Music with no distortion sounds like an authentic version of what was performed.

Frequency balance

Frequency balance refers to the perceived balance between treble (or higher pitch) and bass (or lower pitch) sounds. Within the research activity, we will ask you to rate whether you are experiencing ‘too much treble’ or ‘too much bass’.

Harshness

Harshness refers to an emphasis of certain sound qualities (often in the treble frequencies or higher pitches) that can feel overwhelming, abrasive, painful, or discomforting. Harsh sounds may sound piercing, screechy, shrill or sharp.

Loudness

Loudness is the perceived “strength” of a sound on a scale from quiet to loud. It is a subjective measure that considers the sound intensity, frequency and duration of sounds. Within the research activity, we ask how frequently the music is ‘too quiet’ or ‘too loud’

Enjoyment

Music enjoyment is the pleasure, satisfaction, or positive feelings you experience while listening to the music. It may include appreciation of musical beauty, expressiveness, structure, creativity, or artistic value.

3. Research Activity

You will hear the orchestra perform 10 music excerpts. They are arranged in pairs, where the orchestra play the same bit of the symphony but in different ways. After each excerpt, please give a score i.e. you will be giving scores 10 times.

We want your honest, individual score for the overall impression as if you were in a concert. Please do not try and analyse how the orchestra is playing to figure out the "right" answer. There are no wrong or right answers.

The first two excerpts are trial runs so you can get familiar with doing the task.

In each response, you will be asked to circle your rating on different scales. For example, at the end of each excerpt, we will ask you to rate your enjoyment:

How much did you enjoy the excerpt? Please circle a number from 1-10:

not at all enjoyable									extremely enjoyable
1	2	3	4	5	6	7	8	9	10

If you moderately enjoyed listening to the excerpt, you might circle 6:

How much did you enjoy the excerpt? Please circle a number from 1-10:

not at all enjoyable									extremely enjoyable
1	2	3	4	5	(6)	7	8	9	10

We will also ask you to write a sentence or two about your experience:

In 1-2 sentences, what was your overall impression of the musical excerpt:

Please write as clearly as possible so we can accurately record your impressions.

There will be a pause after each musical excerpt for a period of 1 minute for you to complete your responses. A timer will be displayed on the big screen during this time. Note that there is limited time for providing an answer.

If you make a mistake, please scribble through the circle and/or place an X to indicate this is incorrect.

Excerpt 1

Please score on **loudness**.

Loudness is the perceived "strength" of a sound on a scale from quiet to loud. It is a subjective measure that considers the sound intensity, frequency and duration of sounds. Within the research activity, we ask how frequently the music is 'too quiet' or 'too loud'

How often was the excerpt **too quiet**? Please circle a number from 1-10:

never always

1 2 3 4 5 6 7 8 9 10

How often was the excerpt **too loud**? Please circle a number from 1-10:

never always

1 2 3 4 5 6 7 8 9 10

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt? Please circle a number from 1-10:

not at all enjoyable extremely enjoyable

1 2 3 4 5 6 7 8 9 10

Excerpt 2

Please score on **loudness**.

Loudness is the perceived "strength" of a sound on a scale from quiet to loud. It is a subjective measure that considers the sound intensity, frequency and duration of sounds. Within the research activity, we ask how frequently the music is 'too quiet' or 'too loud'.

How often was the excerpt **too quiet**? Please circle a number from 1-10:

never always

1 2 3 4 5 6 7 8 9 10

How often was the excerpt **too loud**? Please circle a number from 1-10:

never always

1 2 3 4 5 6 7 8 9 10

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt? Please circle a number from 1-10:

not at all enjoyable extremely enjoyable

1 2 3 4 5 6 7 8 9 10

Excerpt 3

Please score on **clarity**.

Clarity refers to how well you can hear and distinguish between the different instruments and elements within the music. **Unclear** music may sound indistinct, mushy, or muddy, **clear music** may sound clean and distinct.

How was the excerpt's **clarity**? Please circle a number from 1-10:

very unclear									very clear
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt? Please circle a number from 1-10:

not at all enjoyable									extremely enjoyable
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

Excerpt 4

Please score on **clarity**.

Clarity refers to how well you can hear and distinguish between the different instruments and elements within the music. **Unclear** music may sound indistinct, mushy, or muddy, **clear music** may sound clean and distinct.

How was the excerpt's **clarity**? Please circle a number from 1-10:

very unclear									very clear
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt? Please circle a number from 1-10:

not at all enjoyable									extremely enjoyable
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

Excerpt 5

Please score on **loudness** and **distortion**

Loudness is the perceived "strength" of a sound on a scale from quiet to loud. It is a subjective measure that considers the sound intensity, frequency and duration of sounds. Within the research activity, we ask how frequently the music is 'too quiet' or 'too loud'.

How often was the excerpt **too quiet**? Please circle a number from 1-10:

never									always
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

How often was the excerpt **too loud**? Please circle a number from 1-10:

never									always
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

Distortion can be caused by artefacts that should not be there or do not feel right, e.g., noise, hiss, pops or crackles. It can also be caused by pitches sounding wrong. Music with no distortion sounds like an authentic version of what was performed.

How distorted was the excerpt? Please circle a number from 1-10:

not distorted									very distorted
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt? Please circle a number from 1-10:

not at all enjoyable									extremely enjoyable
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

Excerpt 6

Please score on **loudness** and **distortion**

Loudness is the perceived "strength" of a sound on a scale from quiet to loud. It is a subjective measure that considers the sound intensity, frequency and duration of sounds. Within the research activity, we ask how frequently the music is 'too quiet' or 'too loud'.

How often was the excerpt **too quiet**? Please circle a number from 1-10:

never									always
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

How often was the excerpt **too loud**? Please circle a number from 1-10:

never									always
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

Distortion can be caused by artefacts that should not be there or do not feel right, e.g., noise, hiss, pops or crackles. It can also be caused by pitches sounding wrong. Music with no distortion sounds like an authentic version of what was performed.

How distorted was the excerpt? Please circle a number from 1-10:

not at all distorted									very distorted
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt? Please circle a number from 1-10:

not at all enjoyable									extremely enjoyable
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

Excerpt 7

Please score on **frequency balance** and **harshness**

Frequency balance refers to the perceived balance between treble (or higher pitch) and bass (or lower pitch) sounds. Within the research activity, we will ask you to rate whether you are experiencing 'too much treble' or 'too much bass'.

How was the **treble**? Please circle a number from 1-10:

too little treble									too much treble
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

How was the **bass**? Please circle a number from 1-10:

too little bass									too much bass
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

Harshness refers to an emphasis of certain sound qualities (often in the treble frequencies or higher pitches) that can feel overwhelming, abrasive, painful, or discomforting. Harsh sounds may sound piercing, screechy, shrill or sharp.

How harsh was the excerpt? Please circle a number from 1-10:

not at all harsh									very harsh
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt? Please circle a number from 1-10:

not at all enjoyable									extremely enjoyable
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

Excerpt 8

Please score on **frequency balance** and **harshness**

Frequency balance refers to the perceived balance between treble (or higher pitch) and bass (or lower pitch) sounds. Within the research activity, we will ask you to rate whether you are experiencing 'too much treble' or 'too much bass'.

How was the **treble**? Please circle a number from 1-10:

too little treble									too much treble
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

How was the **bass**? Please circle a number from 1-10:

too little bass									too much bass
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

Harshness refers to an emphasis of certain sound qualities (often in the treble frequencies or higher pitches) that can feel overwhelming, abrasive, painful, or discomforting. Harsh sounds may sound piercing, screechy, shrill or sharp.

How harsh was the excerpt? Please circle a number from 1-10:

not at all harsh									very harsh
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt? Please circle a number from 1-10:

not at all enjoyable									extremely enjoyable
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

Excerpt 9

Please score on **clarity**.

Clarity refers to how well you can hear and distinguish between the different instruments and elements within the music. **Unclear** music may sound indistinct, mushy, or muddy, **clear music** may sound clean and distinct.

How was the excerpt's **clarity**? Please circle a number from 1-10:

very unclear									very clear
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt? Please circle a number from 1-10:

not at all enjoyable									extremely enjoyable
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

Excerpt 10

Please score on **clarity**.

Clarity refers to how well you can hear and distinguish between the different instruments and elements within the music. **Unclear** music may sound indistinct, mushy, or muddy, **clear music** may sound clean and distinct.

How was the excerpt's **clarity**? Please circle a number from 1-10:

very unclear									very clear
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

In 1-2 sentences, what was your overall impression of the musical excerpt:

How much did you enjoy the excerpt? Please circle a number from 1-10:

not at all enjoyable									extremely enjoyable
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

4. Experience-tracking task

The orchestra will now play a full movement of the Symphony.

Please note that this exercise is optional.

This exercise gives you an opportunity to tell us where in a piece of music you experience something noteworthy. This may include positive aspects (e.g., a moment you particularly enjoyed) or any perceived issues (e.g., passages too quiet, distortion, etc). Report on anything that stands out but please do also enjoy the performance.

Note the time displayed on the screen timer (roughly) and describe your experience briefly in a few words. For example, if 30 seconds into the piece you heard distortion in your right hearing aid you might write:

Time	Experience
00:30	Unpleasant distortion in right hearing aid

If 1 minute 20 seconds into the piece you heard a very quiet note:

Time	Experience
00:30	Unpleasant distortion in right hearing aid
01:20	Very quiet note heard which added drama

Please turn over to enter your noteworthy experiences

Time	Experience

Time	Experience

*** End of Research Activity ***